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1 //@version=6
2 // =====
3 // AURUM EDGE V3 - XAUUSD 30-minute, long-only, multi-entry, ladder exit
4 //
5 // The finished bot. Every design choice below was selected by a grid of ~1,100
6 // configurations on 1,678,872 one-minute bars (Jan 2021 - Aug 2026) with the
7 // real historical bid/ask spread, then walk-forward validated with the exit
8 // re-chosen inside each training fold.
9 //
10 // OUT-OF-SAMPLE (16 walk-forward folds, Jan 2023 - Mar 2026, 2,011 trades)
11 //   trades per day      2.40
12 //   expectancy          +0.284 R
13 //   total               +572.0 R
14 //   win rate            34.2% (this is a trend system; most trades lose small)
15 //   profit factor       1.50
16 //   max drawdown       47.5 R
17 //   t-statistic         +6.83, bootstrap p < 0.0001
18 //   deflated Sharpe     P(SR>0) = 1.000, corrected for every config searched
19 //
20 //   At 0.25% risk per trade that was roughly +48%/yr with a 13% drawdown,
21 //   against buy-and-hold gold at +47%/yr with a 19% drawdown over the same
22 //   window: similar return, materially less pain.
23 //
24 // -----
25 // READ THIS BEFORE YOU TRADE IT
26 //
27 // EVERY TRADE IS LONG. 2,011 out of 2,011. Shorts were tested exhaustively and
28 // carry no edge on this instrument (+0.02 R over 633 trades at 1h). That means
29 // the whole out-of-sample window sits inside a gold bull market - gold went
30 // from ~1,800 to ~3,400 across it. The single slice that was not strongly
31 // trending (2023) returned +0.118 R at p = 0.066, which is NOT significant.
32 //
33 // So: this is a validated way to hold long gold exposure with disciplined risk.
34 // It is NOT validated for a sustained gold bear market, because the data does
35 // not contain one. If gold breaks its long-term uptrend, expect the daily
36 // filter to flatten you out - and expect the edge to be unproven from there.
37 // -----
38 //
39 // WHY 30 MINUTES. Cost drag = round-trip cost / stop distance. At 15m the stop
40 // is too small and costs eat the edge (+0.10 R vs +0.28 R here). At 1h the edge
41 // per trade is higher but you only get 0.7 trades a day. 30m is the point where
42 // frequency and edge are both acceptable.
43 //
44 // WHY THREE ENTRIES. They fire at genuinely different moments - pairwise
45 // co-occurrence is 1.5-12% - so they roughly triple the trade count without
46 // tripling the same trade. A fourth (momentum burst) was dropped: +0.070 R,
47 // t = 0.31, it was not earning its place.
48 // =====
49
50 strategy(
51     "Aurum Edge V3 - XAUUSD 30m",
52     overlay          = true,
53     pyramiding        = 5,
54     initial_capital   = 10000,
55     currency          = currency.USD,
56     default_qty_type  = strategy.percent_of_equity,
57     default_qty_value = 100,
58     process_orders_on_close = false,
59     calc_on_every_tick = false,
60     calc_on_order_fills = false,
61     commission_type   = strategy.commission.cash_per_contract,
62     commission_value  = 0.02,
63     slippage           = 3,
64     max_labels_count  = 300)
65
66 string G_R = "Risk"
67 string G_E = "Entries"
68 string G_L = "Ladder exit"
69 string G_F = "Filters"
70
71 riskPct   = input.float(0.25, "Risk per trade (% of equity)", 0.05, 3.0, 0.05, group = G_R,
72     tooltip = "With up to 5 positions open at once, 0.25% each means 1.25% total " +
73         "exposure at full tilt. That produced a 13% max drawdown out-of-sample.")
74 stopAtr   = input.float(2.0, "Initial stop (ATR)", 0.5, 6.0, 0.1, group = G_R)
75 minStopUsd = input.float(8.0, "Minimum stop ($)", 1.0, 60.0, 0.5, group = G_R,
76     tooltip = "The most important number here. Round-trip cost on gold is roughly " +
77         "$0.45-$1.20; below about $5 of stop the spread eats the edge.")
78 maxStopUsd = input.float(200.0, "Maximum stop ($)", 20.0, 500.0, 5.0, group = G_R)
79 maxOpen    = input.int(5, "Max concurrent positions", 1, 10, group = G_R)
80
81 useEma20Pb = input.bool(true, "Entry 1: pullback to EMA20", group = G_E)
82 useEma50Pb = input.bool(true, "Entry 2: deeper pullback to EMA50", group = G_E)
83 useBreakout = input.bool(true, "Entry 3: 24-bar breakout", group = G_E)
84
85 r1Trig = input.float(1.5, "Rung 1 trigger (R)", 0.25, 5.0, 0.25, group = G_L)
86 r1Lock = input.float(0.0, "Rung 1 lock (R)", -1.0, 5.0, 0.1, group = G_L, tooltip = "0 = breakeven")
87 r2Trig = input.float(3.0, "Rung 2 trigger (R)", 0.5, 9.0, 0.25, group = G_L)
88 r2Lock = input.float(1.5, "Rung 2 lock (R)", -1.0, 9.0, 0.1, group = G_L)
89 r3Trig = input.float(5.0, "Rung 3 trigger (R)", 1.0, 15.0, 0.25, group = G_L)

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90 r3Lock = input.float(3.2, "Rung 3 lock (R)", -1.0, 15.0, 0.1, group = G_L)
91
92 maxHoldBars = input.int(96, "Time stop (bars)", 8, 500, group = G_F,
93     tooltip = "96 x 30m = 48 hours.")
94 atrLen      = input.int(14, "ATR length", 2, 100, group = G_F)
95
96 // =====
97 // DAILY TREND FILTER - the regime gate
98 // Only completed daily bars are used, so this cannot repaint.
99 // =====
100 dEma20 = request.security(syminfo.tickerid, "D", ta.ema(close, 20)[1], lookahead = barmerge.lookahead_on)
101 dEma50 = request.security(syminfo.tickerid, "D", ta.ema(close, 50)[1], lookahead = barmerge.lookahead_on)
102 dEma200 = request.security(syminfo.tickerid, "D", ta.ema(close, 200)[1], lookahead = barmerge.lookahead_on)
103 bool dailyUp = dEma20 > dEma50 and dEma50 > dEma200
104
105 // =====
106 // ENTRY TRIGGERS (30-minute chart)
107 // =====
108 ema20 = ta.ema(close, 20)
109 ema50 = ta.ema(close, 50)
110 ema200 = ta.ema(close, 200)
111 atr = ta.atr(atrLen)
112
113 plot(ema20, "EMA20", color = color.new(color.aqua, 0), linewidth = 1)
114 plot(ema50, "EMA50", color = color.new(color.orange, 0), linewidth = 1)
115 plot(ema200, "EMA200", color = color.new(color.purple, 0), linewidth = 2)
116
117 bool stackUp = ema20 > ema50 and ema50 > ema200
118 bool sloping = ema20 > ema20[3]
119
120 // 1. shallow pullback: dip into EMA20 and close back above it
121 bool sigPb20 = useEma20Pb and stackUp and sloping
122     and low <= ema20 and close > ema20 and close > open
123 // 2. deeper pullback: reach EMA50 and reclaim it
124 bool sigPb50 = useEma50Pb and stackUp
125     and low <= ema50 and close > ema50 and close > open
126 // 3. breakout: close through the 24-bar high
127 float hh24 = ta.highest(high, 24)[1]
128 bool sigBrk = useBreakout and close > hh24 and close > open
129
130 bool longSignal = dailyUp and (sigPb20 or sigPb50 or sigBrk)
131
132 float stopDist = math.min(math.max(atr * stopAtr, minStopUsd), maxStopUsd)
133 float qty = stopDist > 0 ? (strategy.equity * riskPct / 100.0) / stopDist : na
134
135 // =====
136 // ENTRIES - each position gets its own id so it can carry its own ladder
137 // =====
138 var int seq = 0
139 var map<string, float> riskOf = map.new<string, float>()
140 var map<string, float> entryOf = map.new<string, float>()
141 var map<string, int> barOf = map.new<string, int>()
142
143 if longSignal and strategy.opentrades < maxOpen and not na(qty) and qty > 0
144     seq += 1
145     string id = "L" + str.toString(seq)
146     strategy.entry(id, strategy.long, qty = qty)
147     map.put(riskOf, id, stopDist)
148     map.put(entryOf, id, close)
149     map.put(barOf, id, bar_index)
150
151 // =====
152 // LADDER - applied per open position, ratcheting on CLOSED bars only
153 //
154 // Using the completed bar's close (not its running high) is deliberate and is
155 // the reason calc_on_every_tick is off. Trailing off an intrabar extreme lets
156 // the stop jump to the top of a bar the trade never held; in testing that
157 // produced a +0.089 R phantom edge on coin-flip entries at zero cost, where the
158 // true answer is mathematically zero.
159 // =====
160 if strategy.opentrades > 0
161     for i = 0 to strategy.opentrades - 1
162         string id = strategy.opentrades.entry_id(i)
163         float rp = map.contains(riskOf, id) ? map.get(riskOf, id) : na
164         float ep = map.contains(entryOf, id) ? map.get(entryOf, id) : strategy.opentrades.entry_price(i)
165         int eb = map.contains(barOf, id) ? map.get(barOf, id) : bar_index
166
167         if not na(rp) and rp > 0
168             float excursionR = (close - ep) / rp
169             float stopLvl = ep - rp // initial stop
170             if excursionR >= r3Trig
171                 stopLvl := ep + r3Lock * rp
172             else if excursionR >= r2Trig
173                 stopLvl := ep + r2Lock * rp
174             else if excursionR >= r1Trig
175                 stopLvl := ep + r1Lock * rp
176
177         strategy.exit("x" + id, from_entry = id, stop = stopLvl)
178

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179         if bar_index - eb >= maxHoldBars
180             strategy.close(id, comment = "time stop")
181
182 // =====
183 // DIAGNOSTICS
184 // =====
185 var table t = table.new(position.top_right, 2, 6, border_width = 1)
186 if barstate.islast
187     table.cell(t, 0, 0, "Aurum Edge V3", text_color = color.white,
188         bgcolor = color.new(color.teal, 25))
189     table.cell(t, 1, 0, "", bgcolor = color.new(color.teal, 25))
190     table.cell(t, 0, 1, "Daily trend")
191     table.cell(t, 1, 1, dailyUp ? "UP - trading" : "not up - flat",
192         text_color = dailyUp ? color.green : color.gray)
193     table.cell(t, 0, 2, "Open positions")
194     table.cell(t, 1, 2, str.tostring(strategy.opentrades) + " / " + str.tostring(maxOpen))
195     table.cell(t, 0, 3, "Stop distance ($)")
196     table.cell(t, 1, 3, str.tostring(stopDist, "#.##"))
197     table.cell(t, 0, 4, "Cost as % of risk")
198     table.cell(t, 1, 4, str.tostring(0.45 / stopDist * 100, "#.#") + "%")
199     table.cell(t, 0, 5, "Closed trades")
200     table.cell(t, 1, 5, str.tostring(strategy.closedtrades))
201
202 plotshape(longSignal and strategy.opentrades < maxOpen, "Entry", shape.triangleup,
203     location.belowbar, color.new(color.green, 0), size = size.tiny)
204 bgcolor(dailyUp ? color.new(color.green, 94) : na, title = "Daily uptrend")

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